

HEURISTIC EVIDENCE FOR THE BRIDGING CONJECTURE

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ABSTRACT. We present heuristic and computational evidence that the Bridging Conjecture (BC) holds and that the exception set $\mathcal{X}$ does not grow beyond the 11 known elements of A243956. Conditional on the Hardy-Littlewood Conjecture (HLC), the probability that the generative sieve stalls at any step tends to zero, and the probability that any integer $w > 701$ fails to be a sum of two twin prime witnesses decreases super-exponentially with w . Figures drawn from computational data to $w = 400,000$ confirm the predicted decay of gap ratios and growth of decomposition counts. We also argue from an additive perspective why BC is overwhelmingly plausible despite the density of twin prime witnesses being zero, and why sieve methods are ill-suited to proving it. This paper assumes familiarity with the generative sieve [1] and the Bridging Conjecture [2].

1. INTRODUCTION

The Bridging Conjecture (BC) [2] asserts: for every $V \in \mathbb{N}$ there exist $u, v, w \in \mathcal{A}_\infty$ with $u \leq v \leq V < w$ and $u + v = w$. It implies the Twin Prime Conjecture (TPC), but no proof is known.

This paper collects the heuristic evidence that BC is true. The evidence comes from three independent directions:

- (1) **The generative sieve perspective:** conditional on HLC, the probability that the sieve stalls (the priority queue empties) at any step tends to zero as the sieve grows. The sieve is defined in [1]; we recall the necessary notation in Section 2.
- (2) **The decomposition count perspective:** the expected number of ways to write a large integer w as a sum $u + v$ with $u, v \in \mathcal{A}_\infty$ grows without bound under HLC, making it overwhelmingly unlikely that any $w > 701$ lies in the exception set $\mathcal{X}$.
- (3) **The additive perspective:** the candidate pair count grows like $V^2/(\log V)^4$ under HLC, and arithmetic correlations in the singular series actively favour witnesses. The obstruction to proof is correlation control, not plausibility.

None of these arguments constitutes a proof. The parity barrier [5] prevents converting probabilistic sieve arguments into unconditional results. BC must

currently be taken as an independent conjecture, further supported by the empirical record extending to $w \approx 3.3 \times 10^9$ by Dubner [3].

2. NOTATION

We recall the key definitions from [1] and [2].

$\mathcal{A}_\infty$ denotes the set of all twin prime witnesses (A002822): integers w such that both $6w - 1$ and $6w + 1$ are prime. $\mathcal{A} \subseteq \mathcal{A}_\infty$ denotes the finite set built by the generative sieve up to some limit.

The generative sieve partitions $\mathbb{N}$ into three classes:

$$\begin{aligned}\mathcal{A} &= \{w : w \text{ was enqueued by the sieve}\}, \\ \mathcal{B} &= \{w : w \text{ was classified but not enqueued}\}, \\ \mathcal{X} &= \mathbb{N} \setminus (\mathcal{A} \cup \mathcal{B}).\end{aligned}$$

When the element w_k with dequeue index k is processed, it pairs with every $u \in \mathcal{A}_k = \{w_0, \dots, w_k\}$, producing $N_k = k + 1$ candidates of magnitude $\sim w_k + \max(\mathcal{A}_k) \leq 2 \max(\mathcal{A}_k)$. Write $M_k = \max(\mathcal{A}_k)$.

The twin prime constant is $C_2 \approx 0.6601618$.

3. THE SIEVE PERSPECTIVE: STALLING PROBABILITY

3.1. Why $\mathcal{B}$ tracks PNT. By the Prime Number Theorem, each of $6w - 1$ and $6w + 1$ is prime independently with probability $\sim 1/\ln(6w)$. The expected number of $\mathcal{B}$ -survivors when processing w_k among N_k candidates is $\sim 2N_k/\ln(6M_k)$ —precisely the PNT prediction $\sim 2w/\ln(6w)$. Computational results confirm this: $\mathcal{B}$ tracks $2w/\ln(6w)$ throughout the range computed.

3.2. The stalling probability.

Theorem 3.1 (Stalling Probability, conditional on HLC). *Conditional on HLC, the probability that processing w_k yields no new twin prime witness tends to zero as $k \rightarrow \infty$:*

$$\Pr(\text{no } \mathcal{A}\text{-survivor when processing } w_k) \approx e^{-2C_2 N_k / (\ln 2M_k)^2} \longrightarrow 0 \quad \text{as } k \rightarrow \infty.$$

Proof. Each of the $N_k = k + 1$ candidates independently passes the twin prime test with probability $p_k \approx 2C_2/(\ln 2M_k)^2$ by HLC. The probability that all fail is $(1 - p_k)^{N_k} \approx e^{-p_k N_k} = e^{-2C_2 N_k / (\ln 2M_k)^2}$. Since $N_k = k + 1$ grows without bound, this probability tends to zero. $\square$

Remark 3.2. The independence assumption fails in general: the parity barrier [5] prevents a rigorous version of this argument. Theorem 3.1 is therefore heuristic.

4. THE DECOMPOSITION COUNT PERSPECTIVE: $\mathcal{X}$ GROWTH PROBABILITY

Theorem 4.1 ($\mathcal{X}$ Growth Probability, conditional on HLC). *Conditional on HLC, the probability that any $w > 701$ fails to be a sum of two elements of $\mathcal{A}_\infty$ —equivalently, that $\mathcal{X}$ grows beyond $\text{A243956} \setminus \{1\}$ —decreases super-exponentially with w .*

Proof. Let $D(w)$ denote the expected number of decompositions of w as $u + v$ with $u, v \in \mathcal{A}_\infty$. Under HLC:

$$D(w) \sim \sum_{j=1}^{w-1} \frac{2C_2}{(\ln j)^2} \cdot \frac{2C_2}{(\ln(w-j))^2} \sim \frac{4C_2^2 w}{(\ln w)^4},$$

which grows without bound. Treating the summands as approximately independent, $\Pr[w \in \mathcal{X}] \approx e^{-D(w)} \approx e^{-4C_2^2 w / (\ln w)^4}$. The probability that $\mathcal{X}$ grows for some $w > W$ is at most

$$\sum_{w>W} e^{-4C_2^2 w / (\ln w)^4},$$

which converges and tends to zero super-exponentially as $W \rightarrow \infty$. $\square$

4.1. Empirical calibration. At $w = 3.3 \times 10^9$ (Dubner’s search limit):

$$D(3.3 \times 10^9) \approx \frac{4 \times (0.6601)^2 \times 3.3 \times 10^9}{(\ln 3.3 \times 10^9)^4} \approx 10^5,$$

giving $\Pr[w \in \mathcal{X}] \approx e^{-10^5}$ —negligible by any measure. Dubner found no exceptions beyond $w = 701$, consistent with this estimate.

5. COMPUTATIONAL EVIDENCE

5.1. Gap ratios. BC is equivalent [2] to the gap bound $w_{n+1} - w_n \leq w_n$ for consecutive elements of $\mathcal{A}_\infty$, or equivalently $\text{gap}(n)/w_n \leq 1$.

Figure 1 plots $\text{gap}(n)/w_{n+1}$ for elements of $\mathcal{A}_\infty$ up to $n = 10,000$. All observed ratios lie well below the bound of 1, and the rolling mean decays steadily toward zero—consistent with the HLC prediction $\text{gap}(n)/w_n \sim (\ln w_n)^2 / (2C_2 w_n) \rightarrow 0$.

5.2. Decomposition gap ratios. Figure 2 plots $r = (v - u)/w$ for every decomposition $u + v = w$ with $u \leq v$ and $u, v \in \mathcal{A}_\infty$, for $w \in \mathcal{A}_\infty \cup \text{A243956}$. Elements of A243956 have no decomposition and are assigned $r = 1$ (grey crosses). Three features support BC and MNC:

- (1) No element of $\mathcal{A}_\infty$ above $w = 1$ is missing a decomposition, supporting MNC (Dubner’s Conjecture 1 [3]).
- (2) The A243956 elements are confined to $w \leq 701$, consistent with AIWBTP.

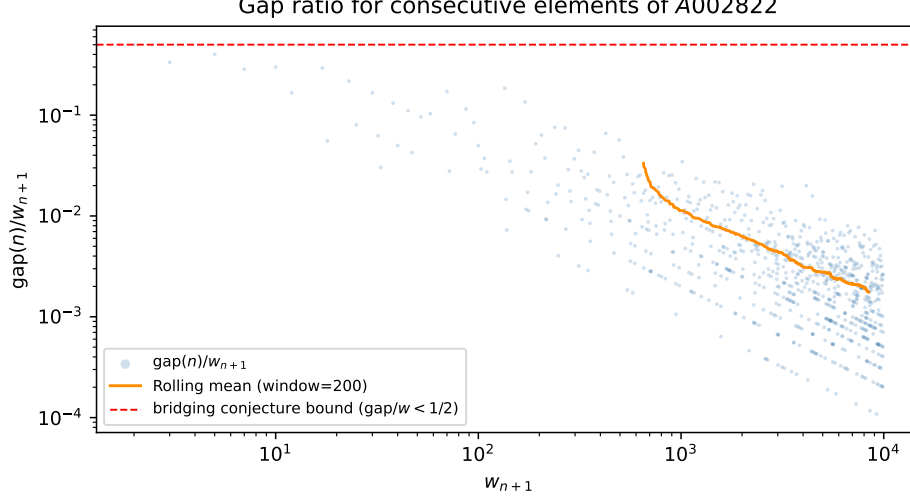


FIGURE 1. Gap ratio $\text{gap}(n)/w_{n+1}$ for consecutive elements of $\mathcal{A}_\infty$ (log scale). The dashed line at $1/2$ marks the bound equivalent to BC. The rolling mean decays toward zero, consistent with the HLC prediction.

- (3) The upper envelope of r trends upward: increasingly lopsided pairs (u, v) with $u \ll v$ become available as w grows, so the sieve always has long-range bridging pairs available.

Figure 3 plots $r_{\min}/r_{\max}$ for elements of $\mathcal{A}_\infty$ with at least two decompositions. A value near 1 would indicate all decompositions are nearly balanced, threatening the gap bound. Instead the ratio trends decisively downward: the spread between most and least balanced decompositions grows with w .

6. THE ADDITIVE PERSPECTIVE

6.1. Why the sieve perspective is ill-suited. The density of $\mathcal{A}_\infty$ is zero: sieve filters leave a surviving density $\prod_p(1 - 2/p) \rightarrow 0$. Viewed through this lens, BC looks hard—it asks for elements in a set of density zero.

The sieve perspective is, however, ill-suited to the question being asked. Sieve methods assign each prime p a discount factor reflecting its long-run asymptotic effect on density. The first time p becomes relevant as a factor of $6m \pm 1$ —near $m \sim p/6$ —is a localised event. The asymptotic factor $2/p$ amortises this event over all m , which is appropriate for counting survivors globally but obscures local arithmetic near any particular V .

6.2. Why the additive perspective favours BC. BC is a local question: for a given V , does some pair (u, v) with $u, v \leq V$ and $u, v \in \mathcal{A}_\infty$ satisfy $u + v > V$ and $u + v \in \mathcal{A}_\infty$?

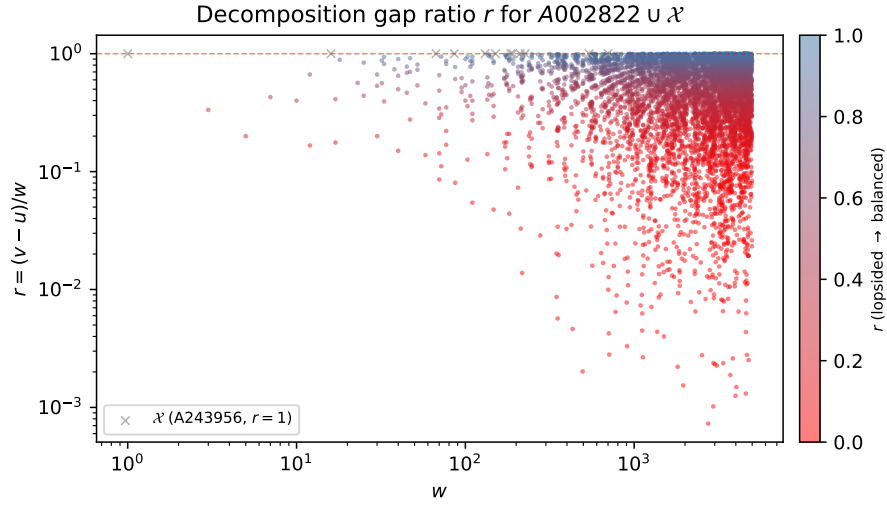


FIGURE 2. Decomposition gap ratio $r = (v - u)/w$ for all decompositions $u + v = w$ with $u, v \in \mathcal{A}_\infty$ (log-log scale). Coloured by r value (red = lopsided, blue = balanced). Grey crosses: A243956 elements, pinned at $r = 1$.

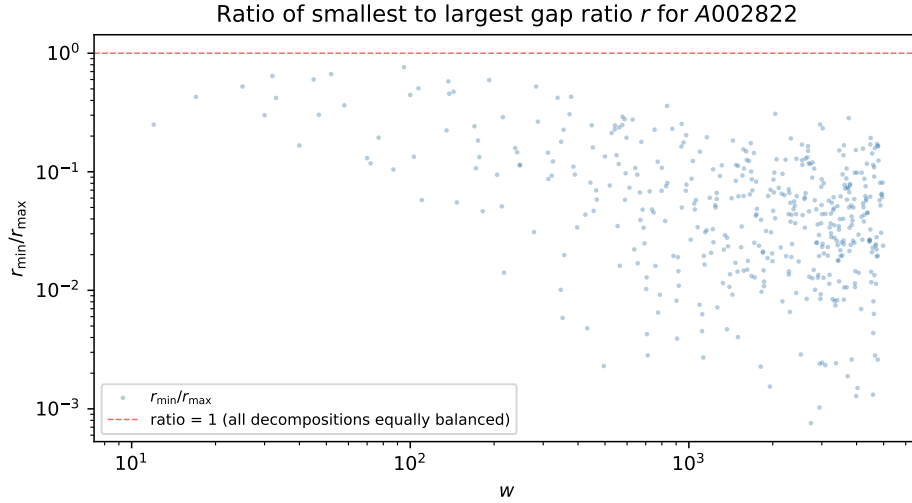


FIGURE 3. Ratio $r_{\min}/r_{\max}$ of smallest to largest decomposition gap ratio for elements of $\mathcal{A}_\infty$ with at least two decompositions (log-log scale). The ratio trends downward, confirming that lopsided bridging pairs remain available as w grows.

The number of candidate pairs is of order $\pi_2(V)^2/2$, growing like $V^2/(\log V)^4$ under HLC. The probability that a given sum $w = u + v \in (V, 2V)$ lies in

$\mathcal{A}_\infty$ is approximately $2C_2/(\log V)^2$, so the expected witness count grows like $V^2/(\log V)^6$ —without bound.

Moreover, the joint sieve constraints on the triple $(u, v, u+v)$ are positively correlated: a residue class that is safe for u and v modulo a prime p tends also to be safe for $u+v$. This means the true singular series constant exceeds the naive $(2C_2)^3$ estimate by a large factor (empirically around 49), so the arithmetic structure actively favours witnesses.

From the additive perspective, BC is not merely plausible but overwhelmingly supported. The remaining gap between heuristic and proof is correlation control: showing that the many candidate witness pairs do not conspire to fail simultaneously. This is the same obstruction that keeps Goldbach’s conjecture open despite its heuristic being overwhelming.

6.3. Why standard additive combinatorics tools do not apply. Freiman’s theorem and the Plünnecke-Ruzsa inequalities apply to sets with *small doubling*: if $|\mathcal{A}+\mathcal{A}| \leq K|\mathcal{A}|$ for some constant K , they give structural information about $\mathcal{A}$. But $\mathcal{A}_\infty$ has *large* doubling: it is sparse with density tending to zero, yet its sumset $\mathcal{A}_\infty + \mathcal{A}_\infty$ covers almost all positive integers beyond 701. The small-doubling hypothesis fails entirely, so these tools have no direct purchase on BC.

Whether the circle method, Fourier analysis on $\mathbb{Z}/N\mathbb{Z}$, or density increment arguments can make progress here that sieve methods cannot is an open question. The framing itself—BC as a local additive statement about a self-reinforcing set—suggests that asymptotic sieve techniques are directed at the wrong structure.

7. LIMITATIONS

Theorems 3.1 and 4.1 are conditional on HLC, which itself remains unproved. The independence assumptions underlying both proofs are not rigorous: the parity barrier [5] is precisely the obstruction that prevents this kind of argument from yielding unconditional results. BC must currently be taken as an independent conjecture, further supported by the empirical evidence that $\mathcal{X}$ has not grown beyond $w = 701$ to the limits of computation.

REFERENCES

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